

Scalability & Future Plan

Date	15 July 2026
Team ID	SWTID-2026-7750
Project Name	Credit Card Approval Prediction System
Maximum Marks	1 Marks

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The application currently uses CSV files instead of a database.	Data management becomes difficult as the number of records increases.	High
2	The Flask development server is suitable only for testing and small-scale usage.	It cannot efficiently handle multiple users simultaneously.	High
3	The system predicts approval based only on the available dataset features.	Prediction accuracy can be improved by including additional financial and credit-related information.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users on the Flask application.	Deploy using a production server with load balancing to support multiple concurrent users.
2	Data Storage	Applicant data is stored in CSV files.	Integrate a relational database such as MySQL or PostgreSQL for efficient data storage and retrieval.
3	Performance	The prediction model works well for small datasets.	Optimize the machine learning model and use caching techniques to improve prediction speed.
4	Security	Basic input validation is implemented.	Add user authentication, secure API communication (HTTPS), and role-based access control to improve security.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy the application on Render and optimize the user interface.	1 Month	Improves accessibility and user experience.
Phase 2	Integrate MySQL or PostgreSQL database for storing applicant information.	2–3 Months	Enhances data management and scalability.
Phase 3	Add user authentication, admin dashboard, and prediction history.	3–4 Months	Improves security and enables better application management.
Phase 4	Implement Explainable AI (SHAP/LIME) and integrate real-time credit score APIs.	5–6 Months	Increases prediction transparency and supports real-world banking applications.